QueryMind AI

Natural Language Database Assistant

System Design Document

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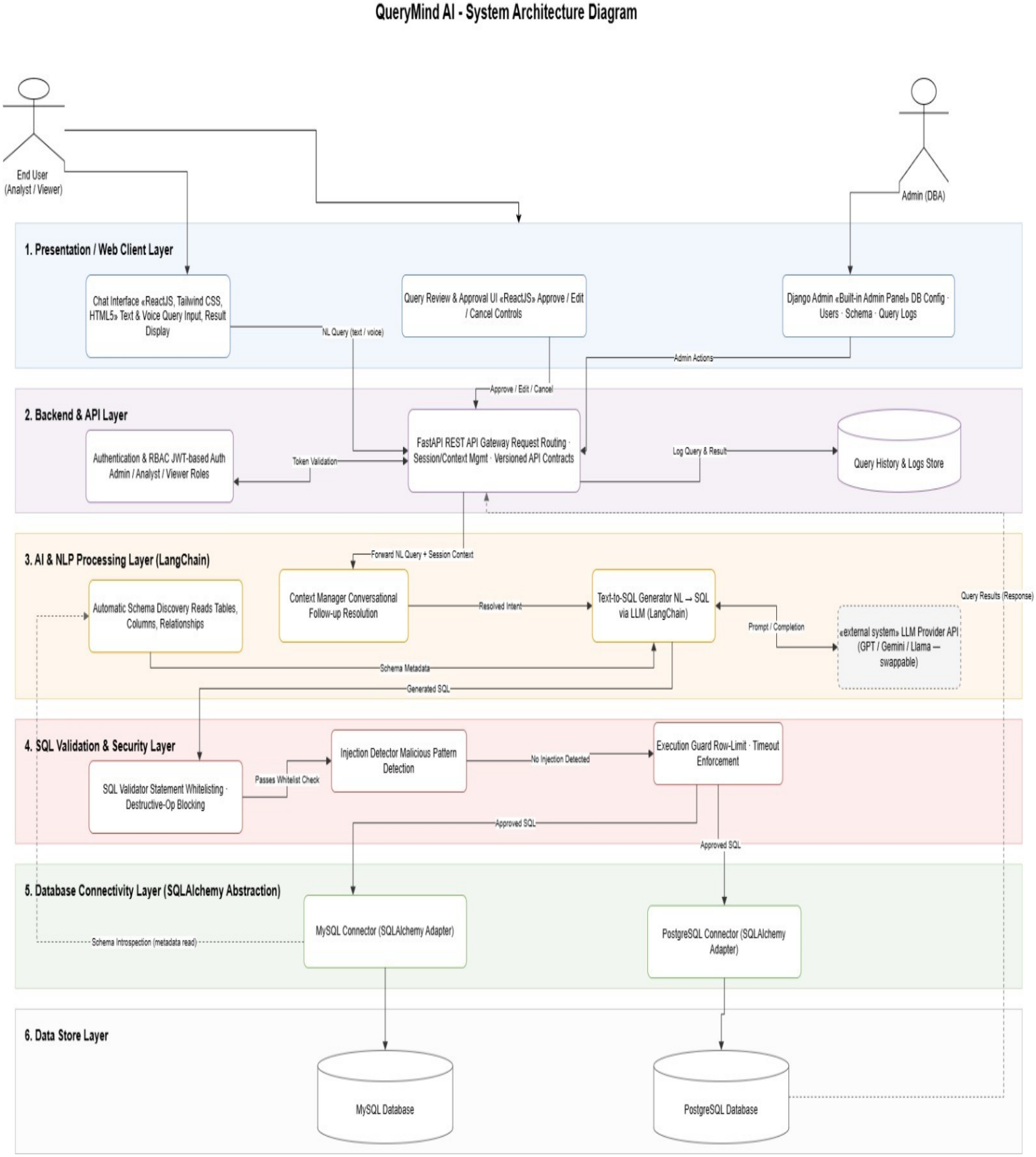
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1. Architecture Diagram



2. Database Schema

1. Users

Field	Type	Required	Description
`_id`	ObjectId	Yes	Unique user identifier
`name`	String	Yes	User's full name
`email`	String	Yes, Unique	Login email
`password_hash`	String	Yes	Hashed password
`role`	String	Yes	`Admin`, `Analyst`, or `Viewer`
`status`	String	Yes	`Active` or `Inactive`
`created_at`	Date	Yes	Account creation time
`updated_at`	Date	Yes	Last profile update

2. Database_Connections

Field	Type	Required	Description
`_id`	ObjectId	Yes	Unique connection identifier
`created_by`	ObjectId	Yes	Admin who created the connection
`name`	String	Yes	Display name of the database
`db_type`	String	Yes	`MySQL` or `PostgreSQL`
`host`	String	Yes	Database host address
`port`	Integer	Yes	Database port
`database_name`	String	Yes	Target database name
`credentials_ref`	String	Yes	Reference to securely stored credentials
`status`	String	Yes	`Connected`, `Disconnected`, or `Error`
`created_at`	Date	Yes	Connection creation time
`updated_at`	Date	Yes	Last status/update time

3. Schema_Access

Field	Type	Required	Description
`_id`	ObjectId	Yes	Unique access-rule identifier
`database_id`	ObjectId	Yes	Reference to `database_connections`
`table_name`	String	Yes	Target table
`column_name`	String	No	Specific column; null means entire table
`allowed`	Boolean	Yes	Whether querying is allowed
`created_at`	Date	Yes	Rule creation time
`updated_at`	Date	Yes	Last modification time

4. Query_Sessions

Field	Type	Required	Description
`_id`	ObjectId	Yes	Unique session identifier
`user_id`	ObjectId	Yes	User who owns the session
`database_id`	ObjectId	Yes	Database used in the session
`title`	String	No	Optional session title
`status`	String	Yes	`Active` or `Closed`
`created_at`	Date	Yes	Session creation time
`updated_at`	Date	Yes	Last activity time

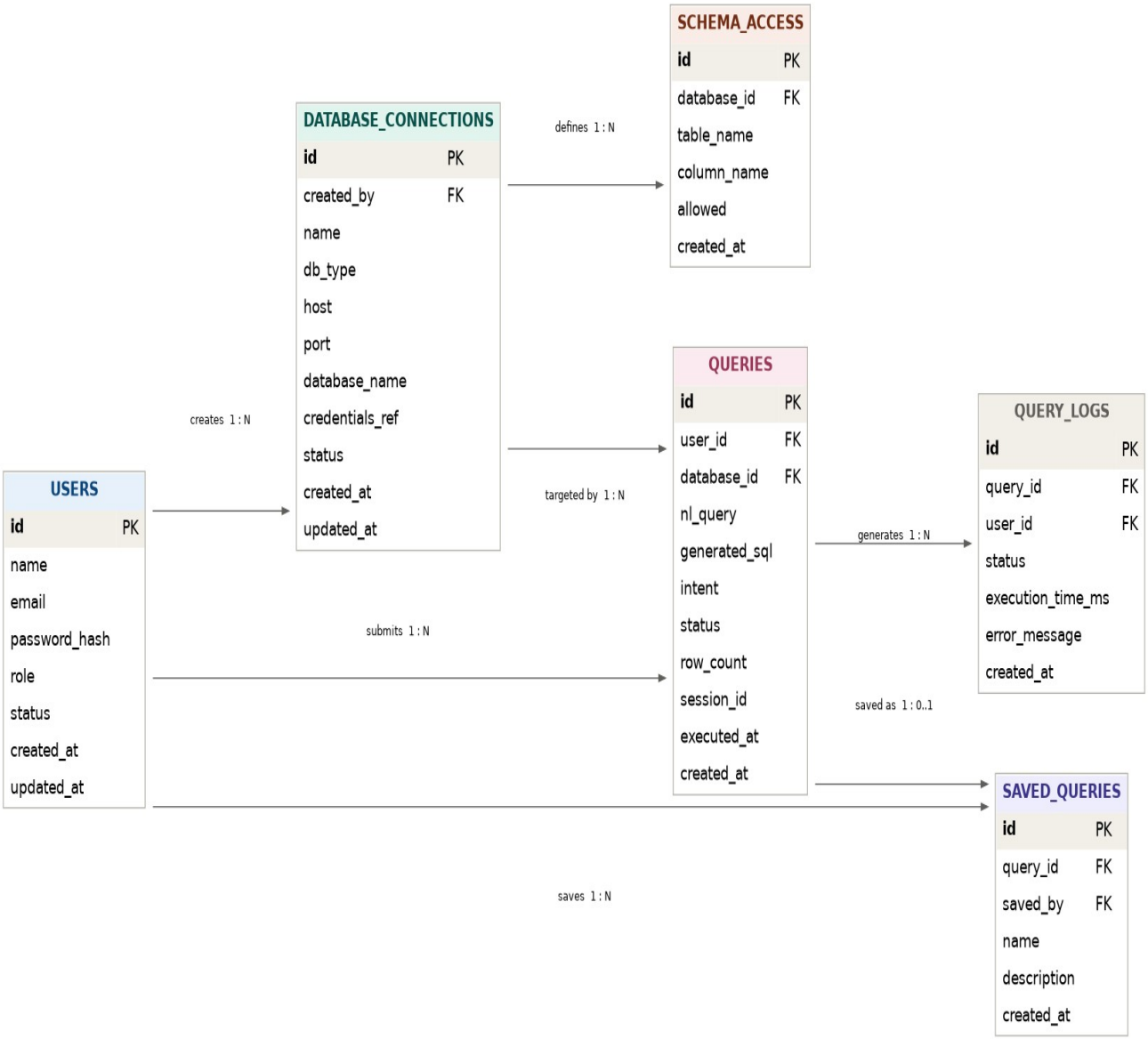
5. Queries

Field	Type	Required	Description
`_id`	ObjectId	Yes	Unique query identifier
`user_id`	ObjectId	Yes	User who submitted the query
`database_id`	ObjectId	Yes	Target database
`session_id`	ObjectId	No	Related conversation session
`nl_query`	String	Yes	Original English/Hinglish query
`input_type`	String	Yes	`Text` or `Voice`
`language`	String	Yes	`English` or `Hinglish`
`intent`	String	No	Query category/intent
`generated_sql`	String	No	SQL generated by the LLM
`validation_status`	String	Yes	`Passed`, `Blocked`, or `Failed`
`validation_message`	String	No	Reason for validation failure/block
`approval_status`	String	Yes	`Pending`, `Approved`, or `Rejected`
`execution_status`	String	Yes	`NotExecuted`, `Success`, `Failed`, or `Timeout`
`row_count`	Integer	No	Number of returned rows
`execution_time_ms`	Integer	No	Query execution time
`executed_at`	Date	No	Execution timestamp
`created_at`	Date	Yes	Query submission time

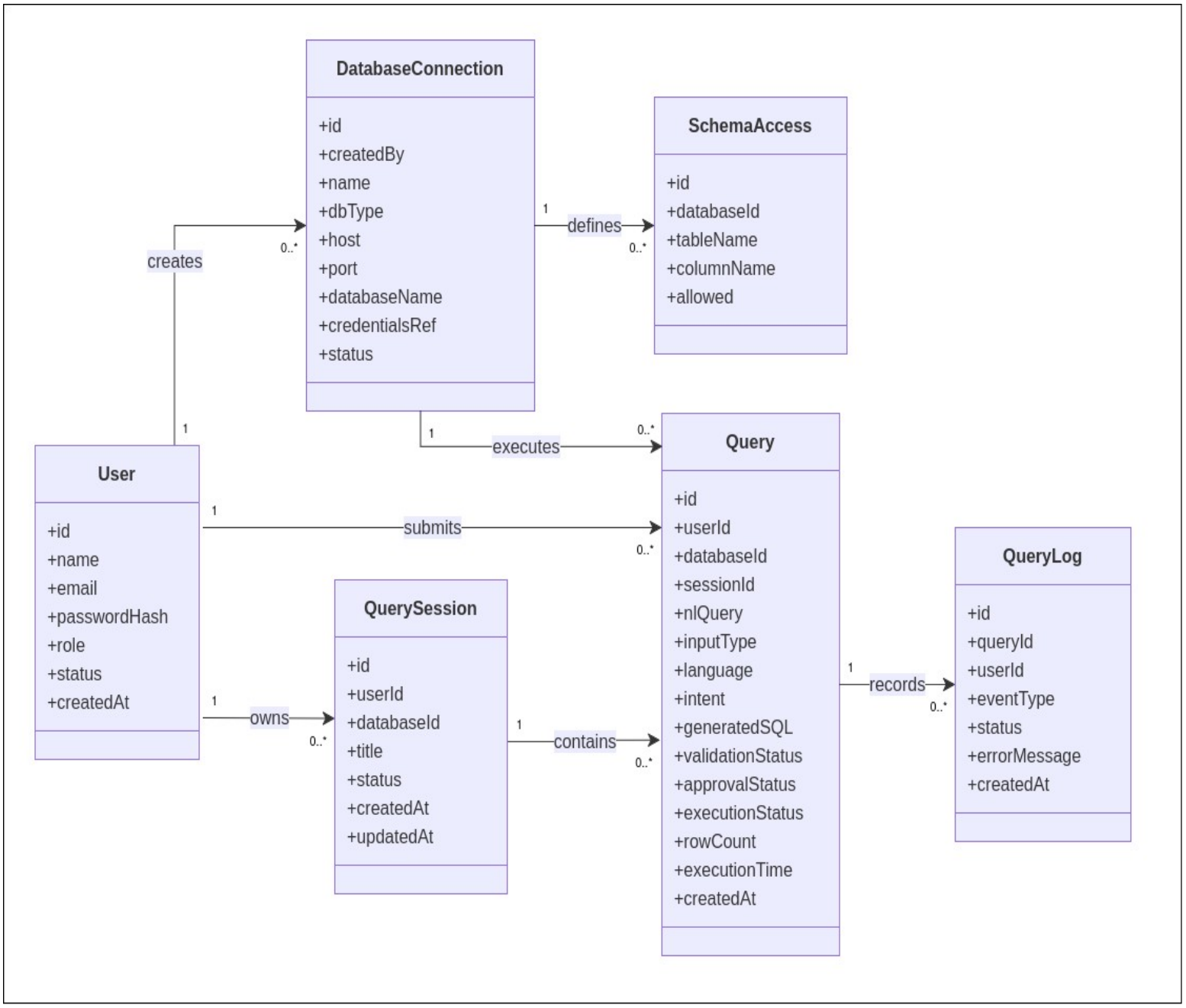
6. Query_logs

Field	Type	Required	Description
`_id`	ObjectId	Yes	Unique log identifier
`query_id`	ObjectId	Yes	Related query
`user_id`	ObjectId	Yes	Related user
`event_type`	String	Yes	Type of query event
`status`	String	Yes	Event result
`error_message`	String	No	User-friendly error message
`created_at`	Date	Yes	Log timestamp

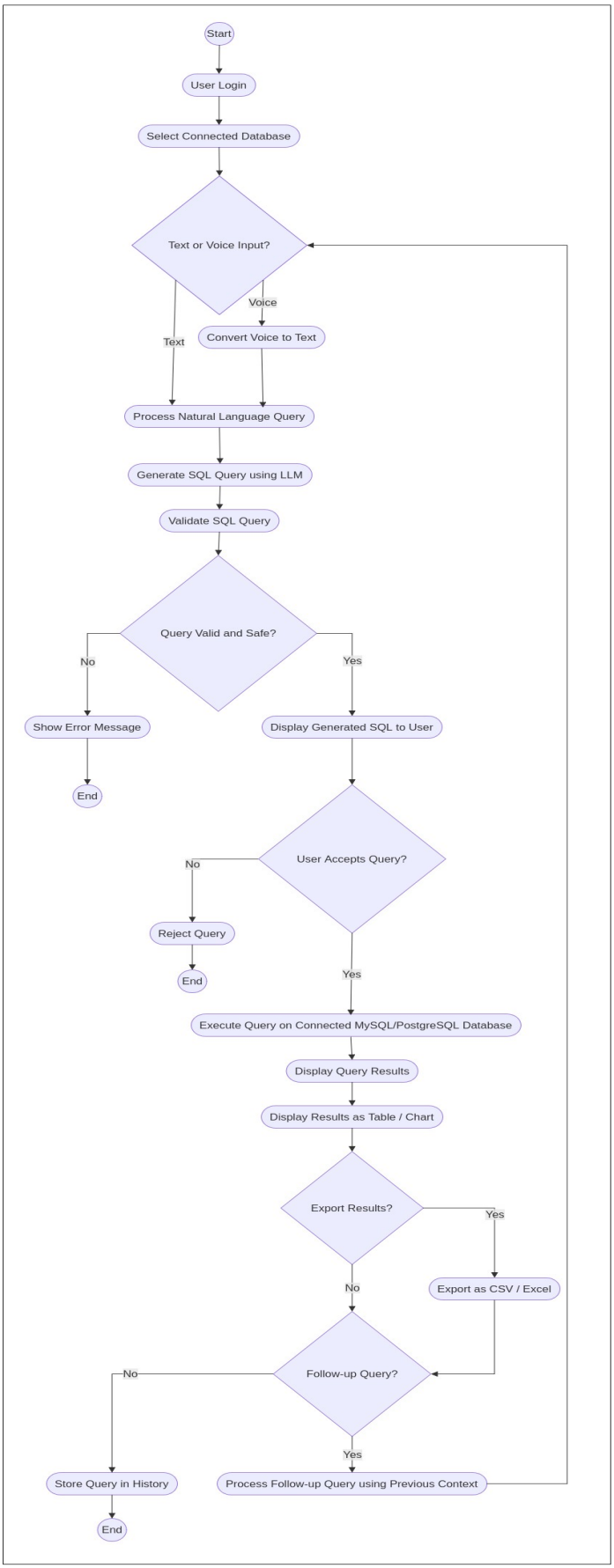
3. ER Diagram


SCHEMA_ACCESS
id PK
database_id FK
table_name
column_name
allowed
created_at
defines 1 : N
DATABASE_CONNECTIONS
id PK
created_by FK
name
db_type
host
port
database_name
credentials_ref
status
created_at
updated_at
creates 1 : N
QUERIES
id PK
user_id FK
database_id FK
nl_query
generated_sql
intent
status
row_count
session_id
executed_at
created_at
targeted by 1 : N
generates 1 : N
QUERY_LOGS
id PK
query_id FK
user_id FK
status
execution_time_ms
error_message
created_at
USERS
id PK
name
email
password_hash
role
status
created_at
updated_at
submits 1 : N
saved as 1 : 0..1
SAVED_QUERIES
id PK
query_id FK
saved_by FK
name
description
created_at
saves 1 : N

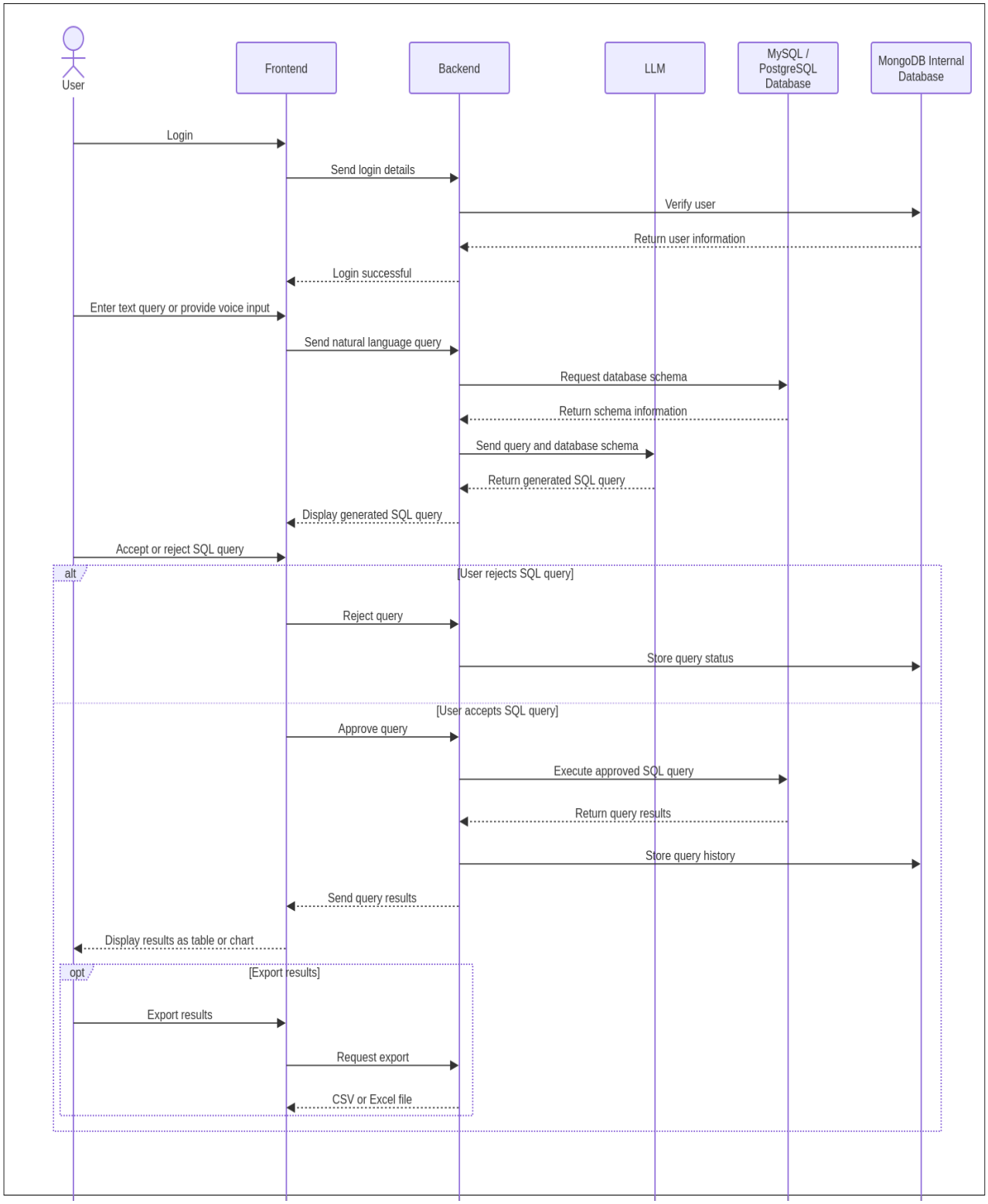
4. Class Diagram



5. Activity Diagram



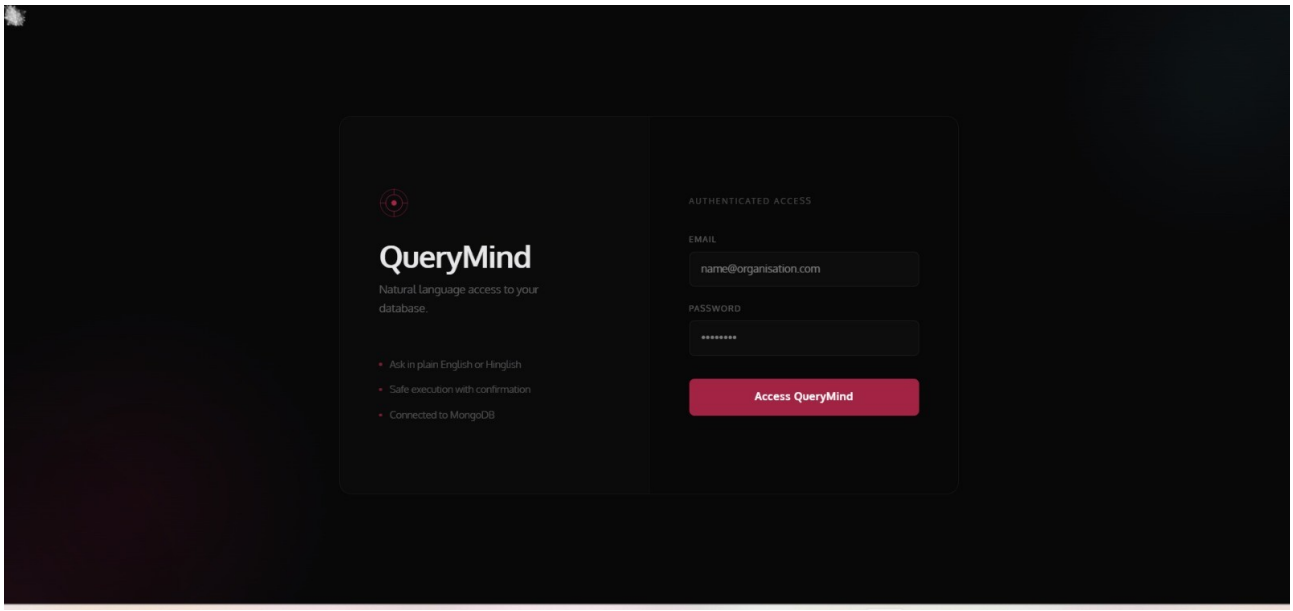
6. Sequence Diagram

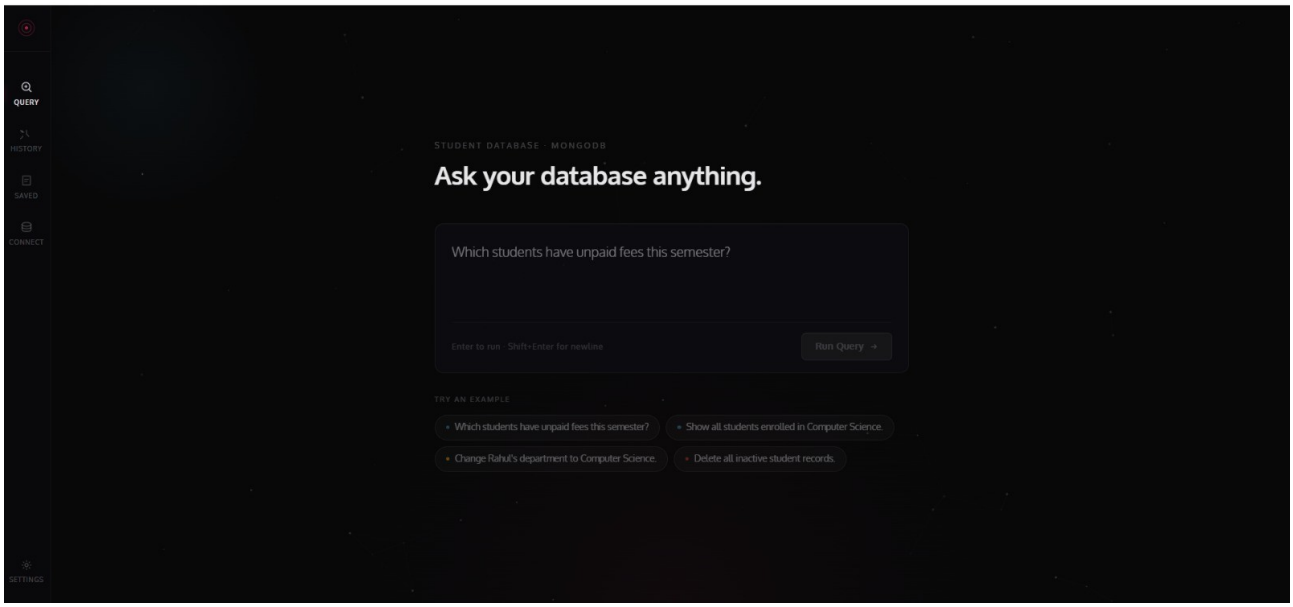


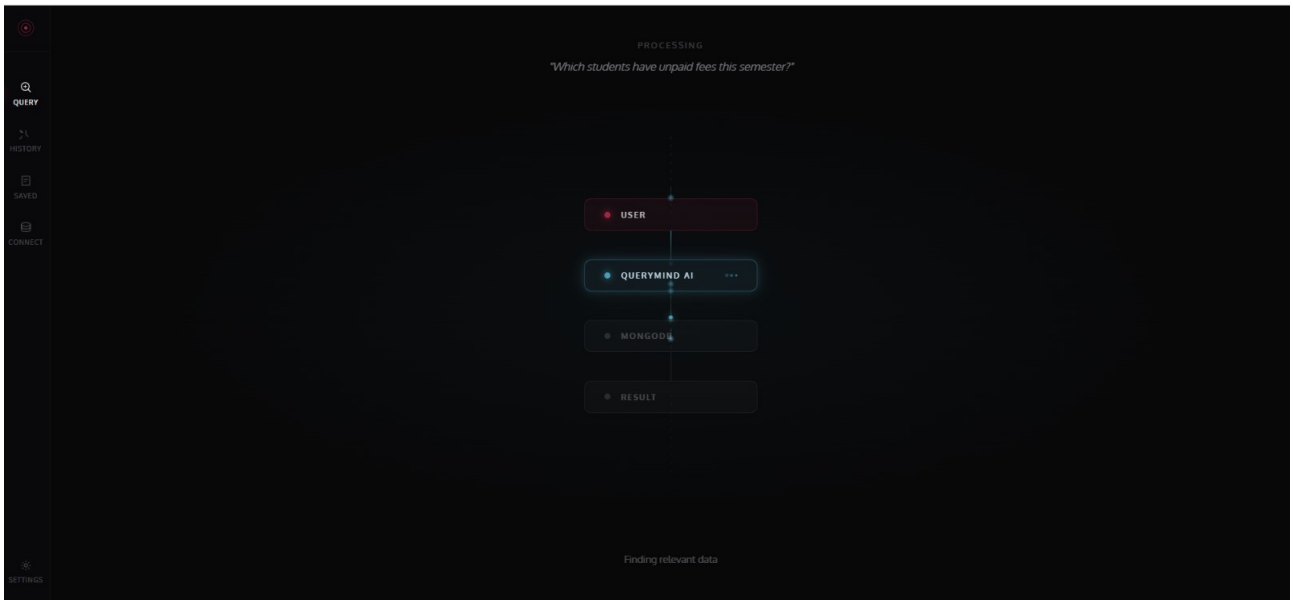
7. REST API Contracts

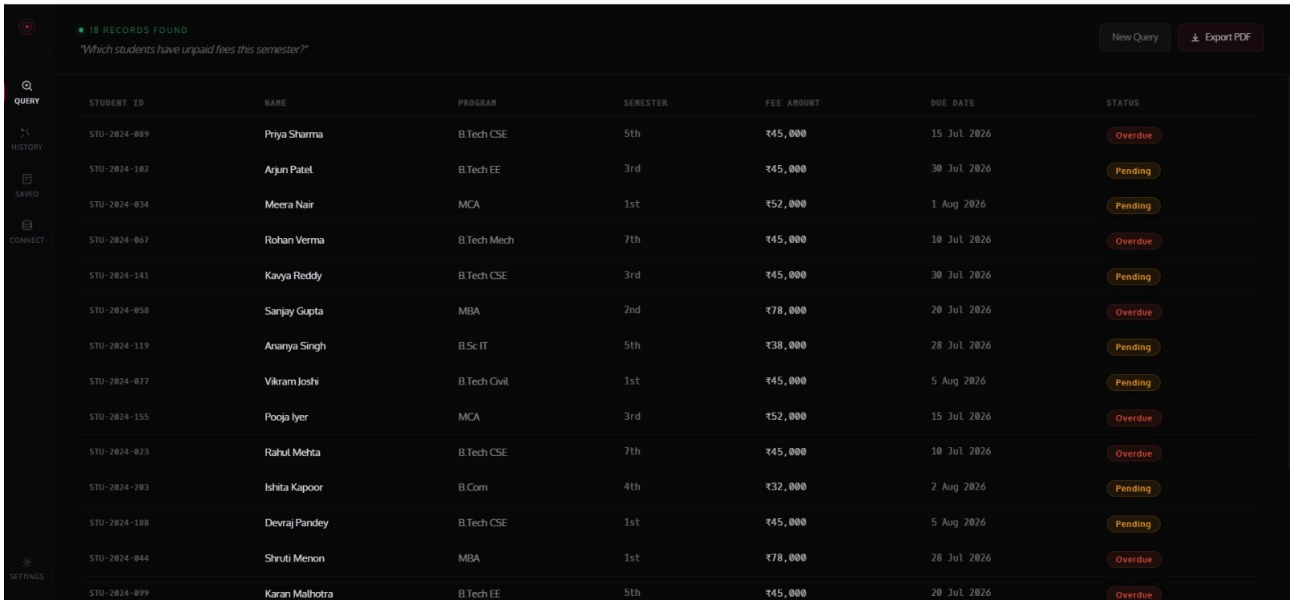
No.	Method	Endpoint	Purpose	Access
1	POST	/api/auth/register	Register a new user	Public
2	POST	/api/auth/login	Login and receive authentication token	Public
3	GET	/api/auth/me	Get logged-in user's profile and role	Authenticated
4	POST	/api/databases	Add/connect a MySQL or PostgreSQL database	Admin
5	GET	/api/databases	View databases available to the user	Authenticated
6	POST	/api/databases/{id}/test	Test whether a database connection works	Admin
7	GET	/api/databases/{id}/schema	Retrieve tables and columns from the connected database	Authorized
8	POST	/api/queries	Submit an English/Hinglish text or voice-converted query and generate SQL	Authenticated
9	GET	/api/queries/{id}	View generated SQL and query status	Authenticated
10	POST	/api/queries/{id}/approve	Approve the generated SQL	Authorized
11	POST	/api/queries/{id}/reject	Reject the generated SQL	Authorized
12	POST	/api/queries/{id}/execute	Execute the approved SQL on the selected database	Authorized
13	GET	/api/queries/{id}/results	Get results of an executed query	Authorized
14	GET	/api/history	View user's previous queries	Authenticated

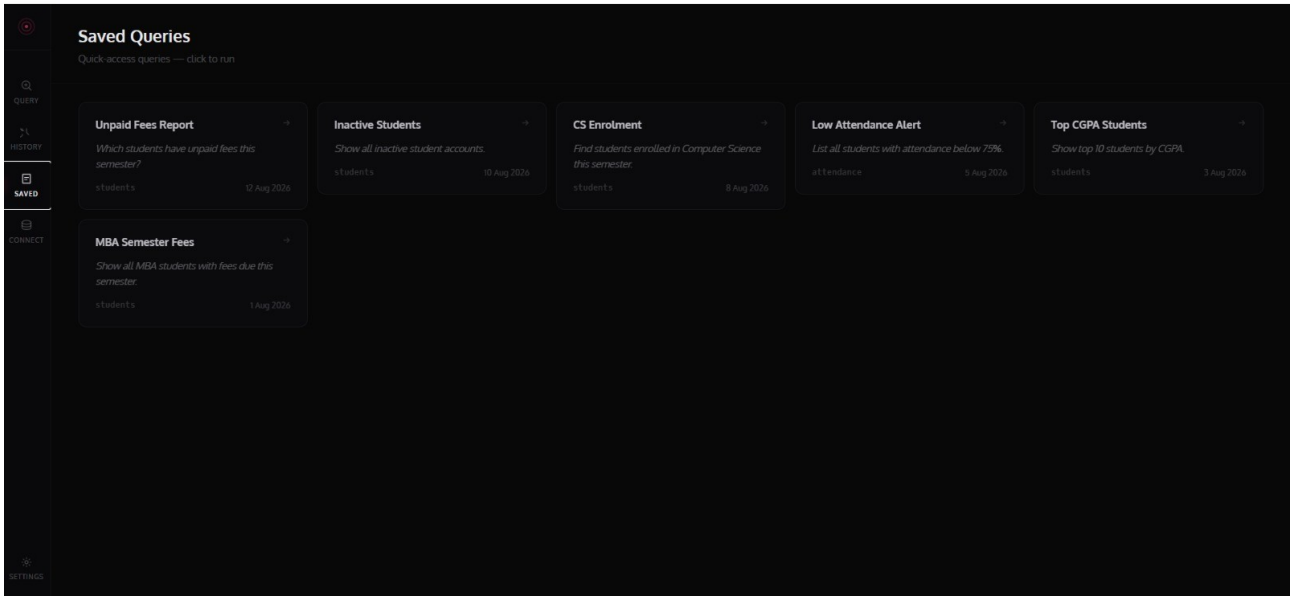
8. UI/UX Mockups

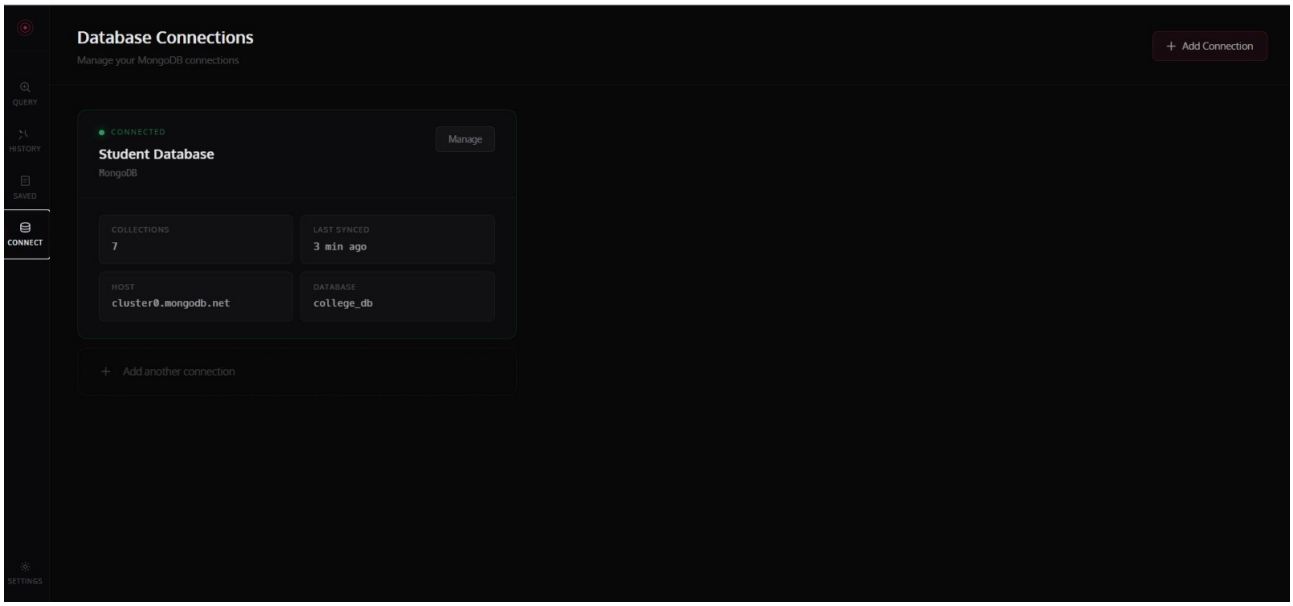





QUERY
HISTORY
SAVED
CONNECT
SETTINGS
PROCESSING
"Which students have unpaid fees this semester?"
USER
QUERYMIND AI
MONGODB
RESULT
Finding relevant data


QUERY
HISTORY
SAVED
CONNECT
SETTINGS
18 RECORDS FOUND
"Which students have unpaid fees this semester?"
New Query
Export PDF
STUDENT ID | NAME | PROGRAM | SEMESTER | FEE AMOUNT | DUE DATE | STATUS
STU-2024-089 | Priya Sharma | B.Tech CSE | 5th | ₹45,000 | 15 Jul 2026 | Overdue
STU-2024-102 | Arjun Patel | B.Tech EE | 3rd | ₹45,000 | 30 Jul 2026 | Pending
STU-2024-034 | Meera Nair | MCA | 1st | ₹52,000 | 1 Aug 2026 | Pending
STU-2024-067 | Rohan Verma | B.Tech Mech | 7th | ₹45,000 | 10 Jul 2026 | Overdue
STU-2024-141 | Kavya Reddy | B.Tech CSE | 3rd | ₹45,000 | 30 Jul 2026 | Pending
STU-2024-058 | Sanjay Gupta | MBA | 2nd | ₹78,000 | 20 Jul 2026 | Overdue
STU-2024-119 | Ananya Singh | B.Sc IT | 5th | ₹38,000 | 28 Jul 2026 | Pending
STU-2024-077 | Vikram Joshi | B.Tech Civil | 1st | ₹45,000 | 5 Aug 2026 | Pending
STU-2024-155 | Pooja Iyer | MCA | 3rd | ₹52,000 | 15 Jul 2026 | Overdue
STU-2024-023 | Rahul Mehta | B.Tech CSE | 7th | ₹45,000 | 10 Jul 2026 | Overdue
STU-2024-203 | Ishita Kapoor | B.Com | 4th | ₹32,000 | 2 Aug 2026 | Pending
STU-2024-188 | Devraj Pandey | B.Tech CSE | 1st | ₹45,000 | 5 Aug 2026 | Pending
STU-2024-044 | Shruti Menon | MBA | 1st | ₹78,000 | 28 Jul 2026 | Overdue
STU-2024-099 | Karan Malhotra | B.Tech EE | 5th | ₹45,000 | 20 Jul 2026 | Overdue


Saved Queries
Quick-access queries — click to run
QUERY
HISTORY
SAVED
CONNECT
SETTINGS
Unpaid Fees Report
Which students have unpaid fees this semester?
students
12 Aug 2026
Inactive Students
Show all inactive student accounts.
students
10 Aug 2026
CS Enrolment
Find students enrolled in Computer Science this semester.
students
8 Aug 2026
Low Attendance Alert
List all students with attendance below 75%.
attendance
5 Aug 2026
Top CGPA Students
Show top 10 students by CGPA.
students
3 Aug 2026
MBA Semester Fees
Show all MBA students with fees due this semester.
students
1 Aug 2026


Database Connections
Manage your MongoDB connections
+ Add Connection
QUERY
HISTORY
SAVED
CONNECT
SETTINGS
CONNECTED
Student Database
MongoDB
Manage
COLLECTIONS
7
LAST SYNCED
3 min ago
HOST
cluster0.mongodb.net
DATABASE
college_db
+ Add another connection